(ii) Distribution function

By definition the distribution function $F(x) = P(X \le x) = \int_{-\infty}^{x} f(u)\,du$

When $x \le 0$ $\quad F(x) = \int_{-\infty}^{x} f(u)\,du = \int_{-\infty}^{x} 0\,du = 0$.

When $x > 0$ $\quad F(x) = \int_{-\infty}^{x} f(u)\,du = \int_{-\infty}^{0} 0\,du + \int_{0}^{x} 2e^{-2u}\,du = 2\left(\frac{e^{-2u}}{-2}\right)_0^x = 1 - e^{-2x}$.

This gives $\quad F(x) = \begin{cases} 0, & \text{for } x \le 0 \\ 1 - e^{-2x} & \text{for } x > 0 \end{cases}$.

(iii) $P(X < 2) = P(X \le 2) = F(2) = 1 - e^{-2\times 2} = 1 - e^{-4}$ (since $F(x)$ is continuous).

(iv) The probability that X is at least equal to four unit of time is
$P(X \ge 4) = 1 - P(X < 4) = 1 - F(4) = 1 - (1 - e^{-2\times 4}) = e^{-8}$.

(v) In the continuous case, $f(x)$ at $x = a$ is not the probability that X takes the value a, that is $f(x)$ at $x = a$ is not equal to $P(X = a)$. If X is continuous type, $P(X = a) = 0$ for $a \in \mathbb{R}$. Therefore $P(X = 3) = 0$.

EXERCISE11.3

1. The probability density function of X is given by $f(x) = \begin{cases} k\,x\,e^{-2x} & \text{for } x > 0 \\ 0 & \text{for } x \le 0 \end{cases}$.

   Find the value of k.

2. The probability density function of X is $f(x) = \begin{cases} x & 0 < x < 1 \\ 2 - x & 1 \le x < 2 \\ 0 & \text{otherwise} \end{cases}$.

   Find (i) $P(0.2 \le X < 0.6)$ (ii) $P(1.2 \le X < 1.8)$ (iii) $P(0.5 \le X < 1.5)$

3. Suppose the amount of milk sold daily at a milk booth is distributed with a minimum of 200 litres and a maximum of 600 litres with probability density function

   $f(x) = \begin{cases} k & 200 \le x \le 600 \\ 0 & \text{otherwise} \end{cases}$

   Find (i) the value of k (ii) the distribution function

   (iii) the probability that daily sales will fall between 300 litres and 500 litres?

4. The probability density function of X is given by $f(x) = \begin{cases} k\,e^{-\frac{x}{3}} & \text{for } x > 0 \\ 0 & \text{for } x \le 0 \end{cases}$

   Find (i) the value of k (ii) the distribution function (iii) $P(X < 3)$

   (iv) $P(5 \le X)$ (v) $P(X \le 4)$.

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